

Diseases and immunity

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Pathogen: Disease-causing organism

Transmissible disease: A disease in which the pathogen can be passed from one host to another.

A pathogen can be transmitted by:

1-Direct contact: Sharing towels, blood transfusion, mixing of bodily fluid

2-Indirect contact: Contaminated surfaces, air, water, food, animals.

Active immunity: A defence against a pathogen by antibody production in the body. It is gained after an infection by a pathogen or a vaccination.

Antibodies: Proteins that bind to antigens leading to direct destruction of pathogens or marking of pathogens for destruction by phagocytes.

→ Each pathogen has its own antigens, which have specific shapes

→ Specific antibodies have complementary shapes which fit specific antigens.

Body defences against infections and pathogens: $\left\{ \begin{array}{l} \text{Chemical barriers} \\ \text{Mechanical barriers} \end{array} \right.$

Mechanical barriers:

Skin: A very tough barrier that has a layer of dead cells and the protein keratin which makes it harder for pathogens to penetrate. Also, the skin has a slightly acidic pH, which makes it harder for pathogens to survive as most prefer neutral pH. It has normal flora, a good bacteria, so not to provide space for pathogens to grow. When skin is wounded, blood clots form to prevent bleeding and pathogens from entering.

Cilia: Hair-like structure in the nose that will catch bacteria and will not allow it to enter.

Chemical barriers:

HCl in the stomach: Kills microbes/pathogens in food.

Mucus in the respiratory tract: traps dust and pathogens. Cilia on the ciliated epithelial cells sweep the mucus away.

Antibodies produced by white blood cells.

Controlling spread of diseases:

- Clean water supply
- Hygienic food preparation
- Good personal hygiene
- Waste disposal
- Sewage treatment.

How do lymphocytes defend the body against pathogens?

When a lymphocyte recognises the antigens on pathogens as foreign antigens, it divides rapidly to form a clone of identical lymphocytes which produce specific antibodies that are complementary to the antigen on the pathogen to destroy the pathogen. Also, part of the clone differentiates into memory cells which provide active immunity and rapid response upon reinfection.

* Memory cells are produced when lymphocytes clone themselves as not all cells produced secrete antibodies against the pathogen. Some lymphocytes will remain in the blood and other parts of the body for a long time.

* In case the pathogen is encountered again, it can be destroyed faster before it is able to cause diseases as the memory cells would be ready to recognise antigens on the pathogens and release complementary and specific antibodies against them. Allowing rapid response upon reinfection and initiating an active immune response.

→ Vaccination involves injecting or ingesting dead or weakened pathogens that do not have the ability to cause disease. Vaccines are weakened pathogens or their antigens that are put into the body, these antigens stimulate an immune response by lymphocytes which produce antibodies. Memory cells are then produced, which give long-term immunity.

→ Vaccinations provide immunity, and mass vaccination programmes can eradicate pathogens as the pathogens will no longer have hosts to infect.

→ Vaccinations to most individuals in a population will reduce the risk of infections for people who cannot be vaccinated due to suppressed immune systems or allergic reactions towards vaccinations.

* Vaccination have helped to control the spread of many diseases:

- Small pox: Which caused scarring and blindness.

- Measles: Which caused brain damage and blindness.

Types of immunity:

1- Active immunity: Long-term immunity against a pathogen by antibodies produced in the body by lymphocytes. This type of immunity develops from acquiring pathogens naturally or by vaccines which contain weakened pathogens. It is long-term because memory cells are produced, but this takes a long time to develop.

2- Passive immunity: Short-term defence against a pathogen by antibodies acquired from another individual. This immunity is acquired as antibodies pass from the mother to child through breast milk or placenta during pregnancy. It is short-term because no memory cells are produced.

* Cholera: A disease caused by a bacterium which is transmitted in contaminated water.

→ Cholera bacterium produces a toxin that causes secretion of chloride ions into the small intestine, causing osmotic movement of water into the gut, causing diarrhoea, dehydration, and loss of ions from the blood.